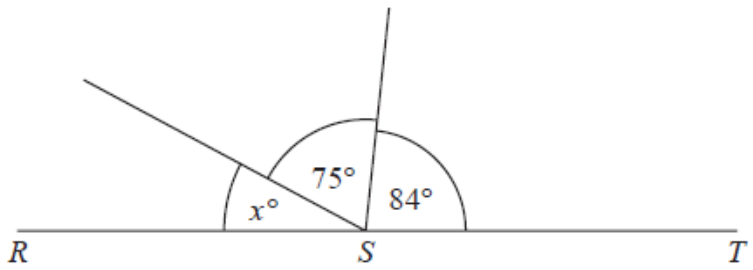
Questions

Q1.



RST is a straight line.

(i) Work out the value of *x*.

.....

(2)

(ii) Give a reason for your answer.

.....

.....

.....

(1)

(Total for question = 3 marks)

Q2.

(a) Work out $\sqrt{\frac{13.82}{4.06}}$

Write down all the figures on your calculator display.

.....

(2)

(b) Give your answer to part (a) correct to 2 decimal places.

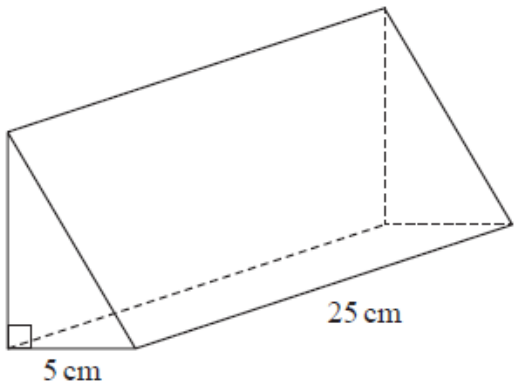
.....

(1)

(Total for question = 3 marks)

Q3.

The diagram shows a prism.



The cross section of the prism is a right-angled triangle.
The base of the triangle has length 5 cm

The prism has length 25 cm
The prism has volume 750 cm^3

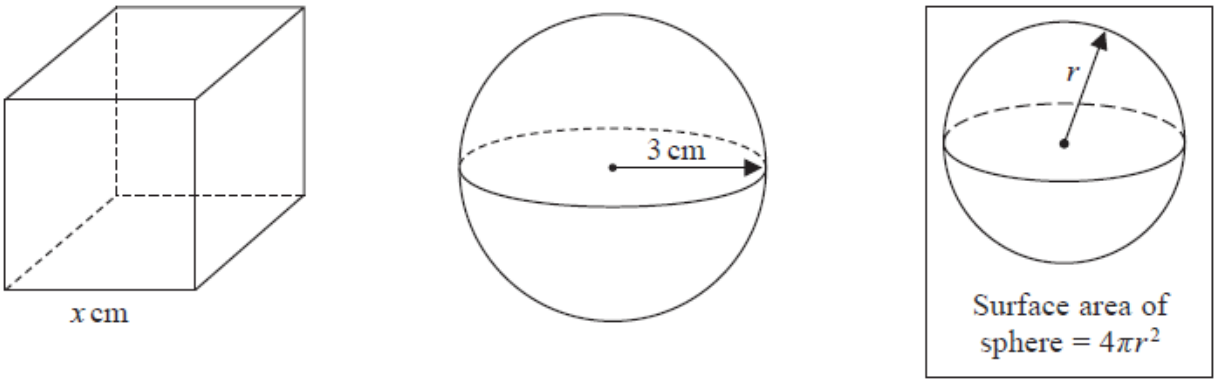
Work out the height of the prism.

..... cm

(Total for question = 3 marks)

Q4.

The diagram shows a cube with edges of length x cm and a sphere of radius 3 cm.



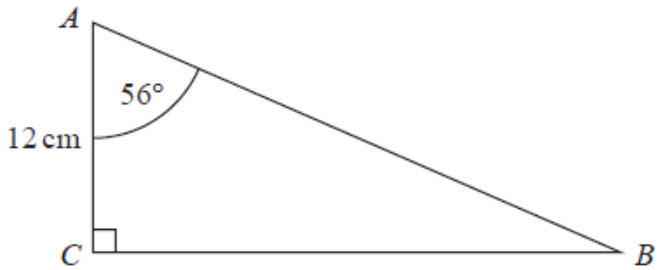
The surface area of the cube is equal to the surface area of the sphere.

Show that $x = \sqrt{k\pi}$ where k is an integer.

(Total for question = 4 marks)

Q5.

ABC is a right-angled triangle.



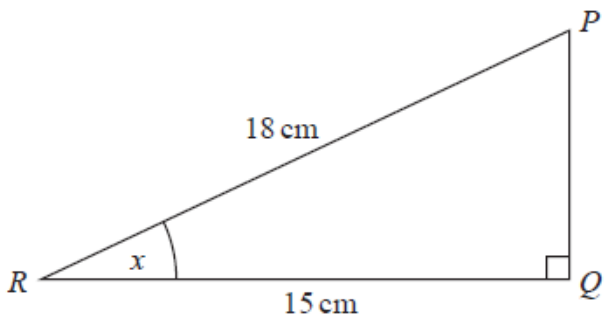
(a) Work out the length of *BC*.

Give your answer correct to 1 decimal place.

..... cm

(2)

PQR is a right-angled triangle.



(b) Work out the size of the angle marked x.

Give your answer correct to 1 decimal place.

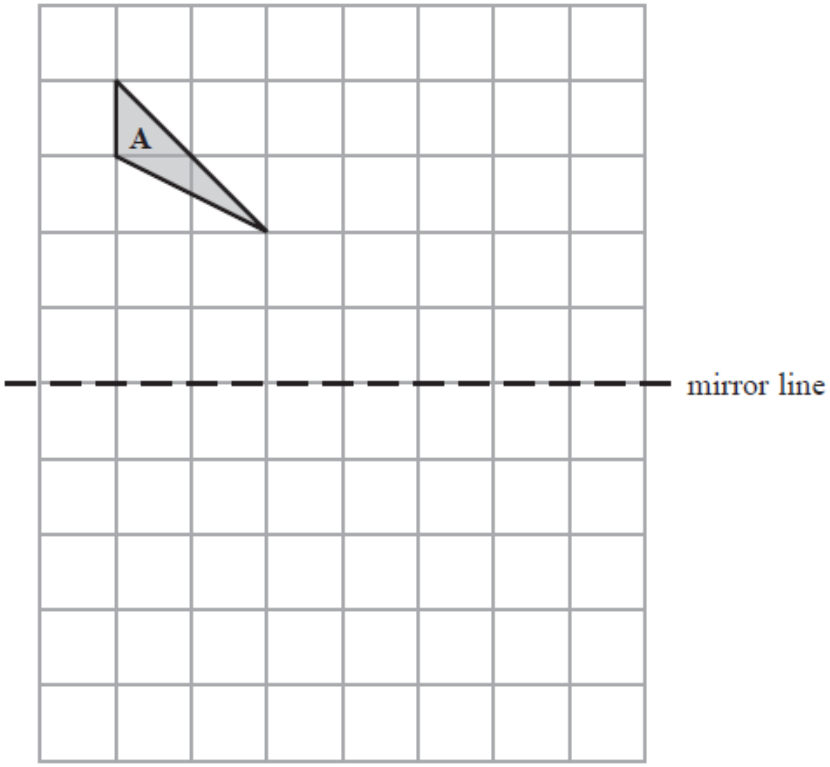
..... °

(2)

(Total for question = 4 marks)

Q6.

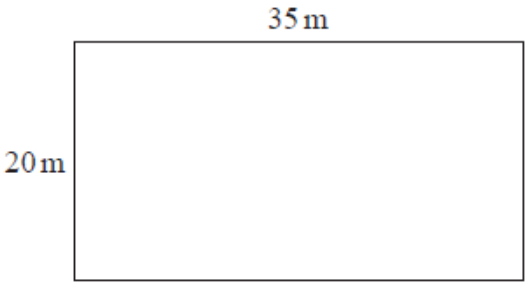
Reflect shape **A** in the mirror line.



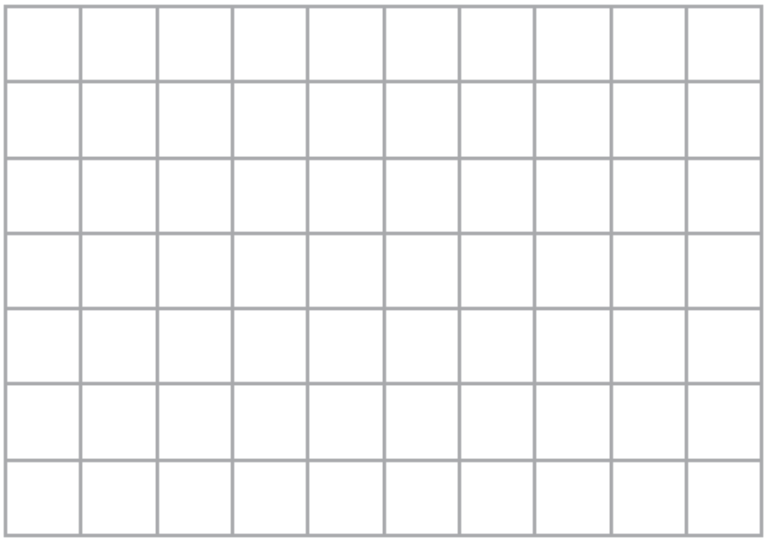
(Total for question = 2 marks)

Q7.

The diagram shows a rectangle.



On the centimetre grid below, draw an accurate scale drawing of this rectangle.
Use a scale of 1 cm to represent 5 m.



(Total for question = 2 marks)

Q8.

(a) Write 196 minutes in hours and minutes.

..... hours ..... minutes

(2)

A train travels *x* miles in 2 hours.

(b) Write down an expression, in terms of *x*, for the average speed of the train.

..... miles per hour

(1)

(Total for question = 3 marks)

Q9.

The equation of a straight line **L** is $y = 3 - 4x$

(i) Write down the gradient of **L**.

.....

(1)

(ii) Write down the coordinates of the point where **L** crosses the y-axis.

(..... , ...)

(1)

(Total for question = 2 marks)

Q10.

Write the following numbers in order of size.
Start with the smallest number.

1.02 0.12 1.20 0.21

.....

(Total for question = 1 mark)

Q11.

Write the following numbers in order of size.
Start with the smallest number.

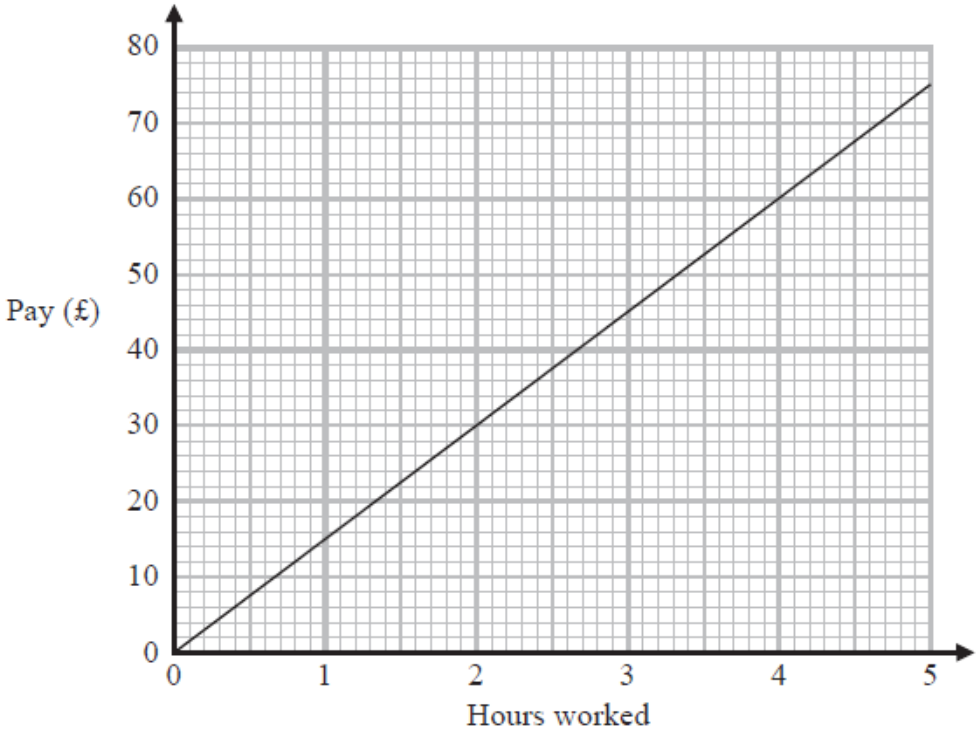
8 –7 –10 1 0 –2

.....

(Total for question = 1 mark)

Q12.

Nazima uses this graph to find out how much money she is paid for the number of hours she has worked.



(a) How much money is Nazima paid for each hour she works?

£ .....

(1)

Last week Nazima worked for 36 hours.

(b) How much money was Nazima paid?

£ .....

(2)

(Total for question = 3 marks)

Q13.

(a) Expand $2(a + d)$

.....

(1)

(b) Factorise $6y^2 - 5y$

.....

(1)

(c) Solve $4x - 7 = 37$

$x =$ .....

(2)

(Total for question = 4 marks)

Q14.

(a) Expand $y(y + 5)$

.....

(1)

(b) Factorise $4a - 6$

.....

(1)

(c) Solve $2(5x - 4) = 21$

$x =$ .....

(3)

(d) Simplify $4e^2f \times 5ef^3$

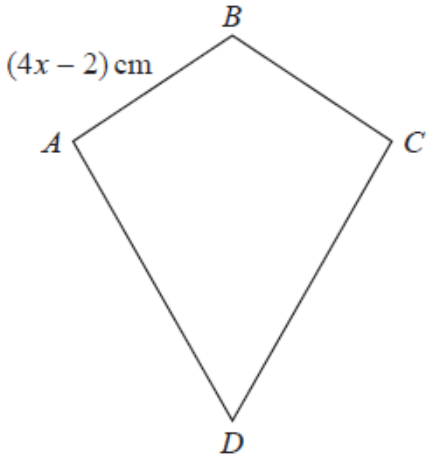
.....

(2)

(Total for question = 7 marks)

Q15.

ABCD is a kite.



AB = (4*x* – 2) cm

Jasper says that *x* could be 0.5

(a) Explain why Jasper cannot be correct.

.....

.....

.....

(1)

AD = 3*AB*
The kite has a perimeter of 64 cm.

(b) Find the value of *x*.

x = ...

(3)

(Total for question = 4 marks)

Q16.

£42 is shared equally between 3 friends.

How much does each friend get?

£ .....

(Total for question = 2 marks)

Examiner's Report

Q1.

The majority of students scored both marks in part (i) calculating the angle to be 21 degrees. A small number of students used 360 in their calculation instead of 180 and this resulted in an incorrect answer of 201 degrees. The other error most often seen was arithmetic, again suggesting that students may have been reluctant to use their calculator for this question.

Part (ii) was less well answered. Students who did score the communication mark often wrote 'angles on a straight line add up to 180'. The mark was most commonly lost for missing out the word angles, other incorrect response included, 'half a circle is 180', here neither angle nor line were mentioned. However, the most common error was for students to state or re state their numerical calculation without mention of geometrical properties. Centres should encourage students to write short succinct explanations.

Q2.

It was pleasing to see that a high proportion of students gained full marks on both parts of this question.

A common incorrect answer to part (a) was 0.9156... from $\frac{\sqrt{13.82}}{4.06}$ i.e., failing to square root the whole calculation, this gained no marks.